

3.1 Factors influencing water availability

Read each scenario in the table. For every scenario, identify **which factor or factors** are influencing water availability in that place, and **explain why**. You may choose from the following factors:

Physical (natural) factors

- Climate (*rainfall, temperature, evaporation, seasons*)
- Geology (*rock type, soil, aquifers, minerals*)
- Topography (*slopes, mountains, basins, elevation*)
- Vegetation (*forests, ground cover, infiltration*)
- Natural drainage patterns (*catchments, inland basins*)

Human factors

- Infrastructure (*pipes, pumps, dams, treatment systems*)
- Demand and water use (*population, farming, industry*)
- Land use (*urbanisation, clearing, agriculture*)
- Management and governance (*control of water, maintenance*)

Scenario	Key factor and why
Scenario 1: A small town in western Queensland receives very little rainfall each year and experiences long periods of high temperatures. Most local creeks only flow after heavy storms, and groundwater refills slowly.	
Scenario 2: A farming community sits on sandstone and gravel, which allow rain to soak into the ground easily. As a result, the region has several healthy aquifers that supply reliable bore water for households and irrigation.	
Scenario 3: A village at the base of a steep mountain range sees water rush downhill quickly during storms. Streams fill rapidly and then dry up just as fast, leaving little time for the water to soak into the soil or refill groundwater.	
Scenario 4: A region that once had thick forest has been heavily cleared for farmland. Rainfall now hits bare soil, causing erosion and muddy runoff into rivers. Water becomes polluted, and less water infiltrates to recharge the groundwater	
Scenario 5: A growing city has increasing demand for water due to more homes, businesses and industry. Even though the region receives moderate rainfall, the main dam struggles to meet the needs of the expanding population.	

Scenario 6: A remote community relies on old pipes and a small water treatment plant that often breaks down. Even when water is available, it is not always safe to drink until repairs are made.	
Scenario 7: In central Australia, water from storms does not flow to the ocean but instead collects in low-lying salt lakes that evaporate quickly. Water is often saline and not suitable for drinking.	
Scenario 8: A dam upstream stores water for agriculture and electricity. Communities downstream often see reduced flow in the river, especially during dry years, limiting their access to freshwater.	
Scenario 9: A farming district uses large amounts of fertiliser. After rain, nutrients wash into local rivers, causing algal blooms that reduce water quality and make treatment more difficult.	
Scenario 10: A small island relies mainly on rainfall collected in tanks and a single freshwater stream. In dry years, water shortages occur quickly because there is little groundwater storage.	

1. Which factor did you notice appeared most often across the scenarios? Why do you think this factor has such a strong influence on water availability in different places?

2. Choose two scenarios that showed very different water availability. Compare them: what physical or human factors caused these differences?
